The Solar System

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Introduction to the Solar System

The Solar System is the Sun together with all the objects that travel around it because of its gravity. This includes eight planets, their moons, dwarf planets, asteroids, comets and clouds of gas and dust. The Solar System formed about 4.6 billion years ago from a giant cloud of gas and dust that collapsed under gravity. The Sun contains about 99.86 percent of the total mass of the whole system. The Solar System lies in the Milky Way galaxy, about 26,000 light-years from the galaxy's centre, and takes roughly 225 to 250 million years to orbit it once.

The Sun

The Sun is a medium-sized star, a huge ball of hot gas mainly of hydrogen and helium. In its core, nuclear fusion converts hydrogen to helium and releases enormous energy. The surface temperature is about 5,500 degrees Celsius and the core is about 15 million degrees Celsius. Sunlight takes about 8 minutes and 20 seconds to reach the Earth, which lies about 150 million kilometres away; this distance is called one astronomical unit (AU).

How Many Planets Are in the Solar System?

There are eight planets in the Solar System. In order from the Sun they are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune. A common memory aid is "My Very Educated Mother Just Served Us Nachos". The planets are divided into two groups. The four inner planets, Mercury, Venus, Earth and Mars, are called terrestrial or rocky planets because they are small with solid surfaces. The four outer planets are much larger: Jupiter and Saturn are gas giants made mostly of hydrogen and helium, while Uranus and Neptune are ice giants made of water, ammonia and methane ices around a rocky core. Between Mars and Jupiter lies the asteroid belt, a region of rocky bodies.

The Planets in Brief

Mercury is the smallest planet and closest to the Sun, with a year of only 88 Earth days. Venus is the hottest planet, with surface temperature about 465 degrees Celsius because of its thick carbon dioxide atmosphere, and it rotates backward compared with most planets. Earth is the only known planet with life and liquid water on its surface, and has one natural satellite, the Moon. Mars, the Red Planet, gets its colour from iron oxide, has the tallest volcano in the Solar System, Olympus Mons, and has been explored by India's Mangalyaan orbiter. Jupiter has a giant storm called the Great Red Spot and dozens of moons, including Ganymede, the largest moon in the Solar System. Saturn is famous for its bright rings made of ice and rock and has well over a hundred known moons. Uranus rotates on its side. Neptune is the farthest planet, with very strong winds.

Which Is the Largest Planet?

Jupiter is the largest planet in the Solar System. Its diameter is about 140,000 kilometres, more than 11 times the diameter of the Earth, and its mass is about 318 times that of the Earth, more than twice the mass of all the other planets combined. More than 1,300 Earths could fit inside Jupiter. Saturn is the second-largest planet. It is a gas giant with no solid surface, and it spins very fast, completing one rotation in about ten hours, which is the shortest day of any planet.

Why Is Pluto Called a Dwarf Planet?

Pluto was discovered in 1930 by the American astronomer Clyde Tombaugh and was regarded as the ninth planet for 76 years. On 24 August 2006 the International Astronomical Union (IAU) defined a planet as a body that orbits the Sun, is massive enough to be nearly round, and has cleared the neighbourhood around its orbit of other objects. Pluto meets the first two conditions but not the third, because its orbit overlaps with the Kuiper Belt and it shares space with many similar icy bodies. Therefore Pluto was reclassified as a dwarf planet. Pluto is about 2,377 km wide, smaller than Earth's Moon, and takes about 248 Earth years to orbit the Sun. NASA's New Horizons spacecraft flew past it in July 2015. Other dwarf planets include Ceres, in the asteroid belt, and Eris, Haumea and Makemake, beyond Neptune.

Small Bodies and Distant Regions

Asteroids are rocky remnants of planet formation, mostly in the asteroid belt. Comets are icy bodies that grow bright tails when they approach the Sun; the famous Halley's Comet appears about every 76 years and will return in 2061. The Kuiper Belt is a region beyond Neptune filled with icy objects, and the far-off Oort Cloud is thought to be the source of long-period comets. Meteoroids that enter the atmosphere create shooting stars called meteors.

Exploration and India's Role

Voyager 1 and Voyager 2, launched in 1977, have travelled beyond the planets into interstellar space. India has explored the Moon with Chandrayaan, Mars with Mangalyaan, and the Sun with Aditya-L1. The Nehru Planetarium in Delhi and the Birla Planetarium in Chennai help students learn about the Solar System.

Challenges and Future

Scientists are searching for life on Mars, and on icy moons such as Europa and Enceladus. NASA's Artemis programme plans to return humans to the Moon, and missions are being planned to reach Mars.

Quick Facts

- There are eight planets: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.
- Jupiter is the largest planet, with a diameter of about 140,000 km.
- Pluto was reclassified as a dwarf planet on 24 August 2006 because it has not cleared its orbit.
- The Solar System is about 4.6 billion years old.
- Sunlight reaches the Earth in about 8 minutes and 20 seconds.